

Chain Crop nexus: Where Agriculture Meets Blockchain

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Abstract: Globalized agriculture supply chains face rising challenges related to food safety, transparency, and authenticity. Traditional centralized systems often struggle to ensure traceability and prevent data manipulation. Blockchain technology offers a decentralized, immutable, and transparent method of recording transactions, enabling stakeholders to verify product origin, growth stages, pricing, and distribution. This paper explores blockchain-based solutions for agricultural traceability, highlighting its benefits in improving trust, eliminating intermediaries, and ensuring fair pricing for farmers. The proposed system demonstrates how blockchain and cryptographic hashing can streamline supply chain operations and build consumer confidence in product quality and safety.

1. INTRODUCTION: Agriculture is a major pillar of India's economy, yet its supply chain suffers from fragmentation, poor documentation, and lack of transparency. Traditional systems rely on paper records and centralized databases that are easily tampered with, leading to issues such as counterfeit seeds, adulterated products, inaccurate data, and weak traceability. These challenges reduce trust among farmers, processors, distributors, retailers, and consumers.

Blockchain technology offers an effective solution by providing a decentralized, tamper-proof, and transparent platform for recording supply chain information. This project proposes a blockchain-based agricultural supply chain system using Binance Smart Chain (BSC) for fast, low-cost transactions and IPFS for secure decentralized storage of farm-related data.

The system enables all stakeholders to record key information—such as seed authenticity, farming practices, storage conditions, processing steps, and distribution details—on a trusted digital ledger. Consumers can access this verified data through QR codes, allowing them to trace the complete journey of the product from farm to table.

By integrating block chain and IPFS, the proposed system aims to create a secure, transparent, and efficient agricultural supply chain that reduces fraud, improves accountability, and builds trust across the entire ecosystem.

2. LITERATURE REVIEW:

M. M. Aung and Y. S. Chang (2014) – "Traceability in a Food Supply Chain: Safety and Quality Perspectives

This research introduces a blockchain-based system for automating food trading and improving trust among stakeholders. Smart contracts are used to handle transactions and enforce agreements without requiring manual verification.

Advantages:

- **Automation with Smart Contracts:** Reduces delays, manual work, and human errors in transactions.
- **Transparent and Tamper-proof Records:** Block chain ensures secure and unchangeable transaction logs.

Disadvantages:

- **Technical Expertise Required:** Farmers and small-scale producers may find blockchain difficult to adopt without training.
- **Scalability Issues:** Performance may slow down if transaction volume increases significantly.

J. Hobbs (2006) – “Liability and Traceability in Agri-Food Supply Chains.

This paper examines how traceability affects accountability and responsibility across agricultural supply chains. The author stresses that proper tracking can help resolve disputes and prevent fraud, protecting both consumers and producers.

Advantages:

- **Clear Responsibility Identification:** Helps determine the exact source of contamination or quality issues.
- **Supports Legal Compliance:** Ensures food industry regulations and safety standards are followed.

Disadvantages:

- **Regulatory Challenges:** Different regions follow different legal frameworks, making universal adoption difficult.
- **Data Standardization Issues:** Lack of uniform data formats complicates traceability across regions.

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3. OBJECTIVES:

- * To implement decentralized traceability for agricultural products.
- * To eliminate intermediaries and support direct farmer–consumer transactions.
- * To enhance transparency and trust in supply chain processes.
- * To ensure secure data sharing using cryptographic hashing mechanisms.

4. METHODOLOGY: The proposed system integrates block chain with a role-based platform that includes buyers, sellers, and administrators. Farmers record product details such as quantity, category, and pricing. Buyers can scan QR codes to verify authenticity and initiate purchase requests. SHA-256 hashing ensures immutability and security of transaction data. Smart contracts automate pricing and approvals while ensuring transparency throughout the product life cycle.

5. RESULTS: The system successfully enables secure product tracking from origin to purchase using QR-based validation. Sellers and buyers can verify pricing, quantity, and transaction history without intermediaries. Testing confirms secure authentication, payment handling, and dynamic crop updates, demonstrating the potential for real-world deployment within agricultural ecosystems.

6. ANALYSIS: Block chain implementation provides significant improvements in data integrity, transparency, and fraud reduction. Farmers gain profit equality, while consumers benefit from access to verified product histories. Although promising, the system requires reliable digital infrastructure, stakeholder adoption, and regulatory support to scale effectively across regional and international food networks.

7. LIMITATIONS:

- * Requires technical infrastructure and digital literacy among farmers.
- * Scalability challenges may arise with increased transactions.
- * Integration with existing logistics systems may require additional interoperability protocols

8. CONCLUSION: Blockchain-enabled agriculture supply chains represent a transformational step toward secure, transparent, and trust-based food ecosystems. By decentralizing control and providing traceability, blockchain improves food safety, reduces counterfeit activity, and promotes fair pricing. With continued development and adoption, blockchain can significantly modernize agricultural logistics and benefit all stakeholders involved.

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